Alexander Detkov

Email: adetkov@caltech.edu
Website: alexanderdetkov.github.io
Phone: +1 (626) 399 5770

Current Academic Research

I develop a mathematical theory of how neural networks—especially transformers—learn *world models*: how *local* observations are glued into a globally consistent model of the dynamics. I study the *topological, geometric, and algebraic structure* that emerges in their representations, and what it reveals about when and how they generalize.

Relevant Publications

2024 **A. Detkov***, A. Bhargava*, C. Witkowski*, M. Thomson. "Prompt Baking." *preprint arXiv:2409.13697*, 2024.
Synopsis: Prompting provides explicit, fine-grained natural-language control, but its influence can decay over long contexts. Weight updates yield durable behavior changes, yet require substantial data and specify objectives only implicitly. Prompt Baking bridges these by converting a prompt into a targeted weight update, training the model to match the prompted distribution. This enables continuous control via "half-baking," improves zero-shot reasoning by baking chain-of-thought examples, supports sequential "knowledge baking" for continual updates, and can further amplify control through an iterative extension called Prompt Pursuit.

2023 **A. Detkov**, M. Salameh, M. Fetrat, J. Zhang, W. Lui, S. Jui, D. Niu. "Reparameterization through Spatial Gradient Scaling." *International Conference on Learning Representations (**ICLR**)*, 2023.
Synopsis: Re-parameterization is a powerful technique used to improve CNN training (e.g. YOLOv7), yet its underlying mechanism remains unknown. We investigate how and why re-parameterization may positively affect model training, as well as propose a new analytical approach for finding high-performance re-parameterizations.

Education

Sep 2024 - Current **Ph.D. in Computation and Neural Systems (CNS)**, Caltech, California, USA.
Advisor: Prof. Matt Thomson. GPA: 4.00/4.00.
Chen Graduate Fellow, Tianqiao & Chrissy Chen Institute for Neuroscience.

Sep 2019 - Apr 2024 **B.S. in Engineering Physics, Minor in Mathematics**, University of Alberta, Canada.
GPA: 4.00/4.00. Graduated first in a class of 648 engineering students.
Governor General's Silver Medal, Dean's Medal in Engineering, and APEGA Past Presidents' Medal in Engineering Physics.

Research Experience

Sep 2024 - Current **World Models in Transformers** | California Institute of Technology
Prof. Matt Thomson
A mathematical theory of world-model learning: how local observations are glued into a globally consistent model of the dynamics, and how topological, geometric, and algebraic structure arises in a model's representations.

- Showed how the structure of the training data governs whether a transformer composes local observations into a globally consistent world model, rather than memorizing them.
- Related the algebraic and geometric structure a model learns to the structure of the underlying state space, and characterized what makes that structure—and the model's generalization—emerge.

May 2023 - Apr 2024 **Explainable Graph Neural Networks** | University of Alberta
Prof. Di Niu
Graph neural networks whose explanations are global and remain faithful under distribution shift.

- Constructed a closed loop between a trained GNN and interpretable symbolic expressions, using function approximation and symbolic regression to extract explanations faithful to the model, then folding that symbolic structure back into training via gradient boosting to improve robustness.
- Studied a policy-gradient formulation of learned graph augmentations as a route to stronger out-of-distribution generalization.

Jan 2022 - Dec 2022 **Neural Architecture Search** | Huawei Research
Prof. Di Niu and Dr. Mohammad Salameh
The learning dynamics of convolutional networks under structural reparameterization, and their use in neural architecture search.

- Initiated and led the direction; proved that structural reparameterization is equivalent to a spatial scaling of gradients, and turned that equivalence into an information-theoretic method that surpassed state-of-the-art reparameterizations at a fraction of their cost, with roughly $2\times$ faster training. First-author paper at ICLR 2023, validated across classification, semantic segmentation, and pose estimation.
- Separately worked on hardware-aware NAS and graph-level compiler optimizations for convolutional and transformer architectures, targeting low-latency inference on mobile neural processing units.

May 2021 - Dec 2021 **Computational Nanofluidics** | University of Alberta
Prof. Wylie Stroberg
The dynamics of single-polymer imbibition into nanocapillaries, studied by molecular and stochastic simulation.

- Built dissipative-particle-dynamics simulations in LAMMPS, and a complementary Langevin simulator in Julia to reach time and length scales beyond molecular dynamics.
- Used stochastic-process theory to establish a maximum in the mean first-passage time of imbibition—a result with direct engineering implications, presented at CSME 2022.

May 2021 - Aug 2021 **Experimental Particle Physics** | University of Alberta and IceCube Observatory
Prof. Juan Pablo Yáñez
Calibration of IceCube digital optical modules from Cherenkov light emitted by atmospheric minimum-ionizing muons.

- Developed statistical methods to filter reconstructed muon tracks by their stochastic energy losses, and released the resulting track-filtering toolkit for use across the collaboration.

Additional Publications

2022 **A. Detkov**, W. Stroberg. "Imbibition of a Single Polymer into a Nanocapillary." *Canadian Society for Mechanical Engineering (CSME)*, 2022.

Fellowships and Awards

2024 Rt. Hon. C. D. Howe Memorial Fellowship ($16,000)
2024 Corporation of the Seven Wardens (Camp 6) Scholarship ($2,000)
2023 NSERC Undergraduate Student Research Award ($6,000)
2023 University of Alberta Undergraduate Scholarship ($2,500)
2023 Louise McKinney Post-Secondary Scholarship ($2,500)
2021 University of Alberta Dean's Research Award ($500)
2021 Joseph and Edwina Charyk Scholarship in Engineering Physics ($2,000)
2020 Louise McKinney Post-Secondary Scholarship ($2,500)
2020 Enbridge Inc Dean of Engineering 2nd Year Scholarship ($1,825)
2019 The Faculty of Engineering Entrance Scholarship ($5,000)

Technical Skills

AI Libraries: PyTorch, PyTorch Geometric, Deep Graph Library, PyReason, TensorFlow, ONNX.
Languages: Python, C, C++, Java, Kotlin, Julia.